Bayram İslamoğlu 150715657 (1)

"On my honor, I have neither given nor received any unauthorized and/or inappropriate assistance for all sessions of this exam. The work done on this exam is totally my own. I understand that by the school code, violation of these principles will lead to a zero grade and is subject to further disciplinary issues"

Bayram İslamoğlu

B

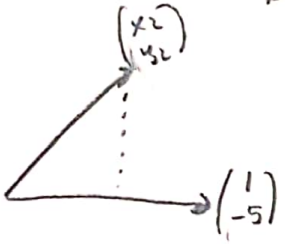
2-) dim K $x - 5y = 0$

a-) $S = \left\{ \begin{pmatrix} 1 \\ -5 \end{pmatrix}, \begin{pmatrix} x_2 \\ y_2 \end{pmatrix} \right\}$ kümesinin $V = R^2$ uzayını germesi gerekir

$R^2 \longrightarrow \begin{pmatrix} a \\ b \end{pmatrix} \in R^2$

$c_1 \cdot \begin{pmatrix} 1 \\ -5 \end{pmatrix} + c_2 \begin{pmatrix} x_2 \\ y_2 \end{pmatrix} = \begin{pmatrix} a \\ b \end{pmatrix}$

b-) $\begin{pmatrix} 1 \\ -5 \end{pmatrix} \begin{pmatrix} x_2 \\ y_2 \end{pmatrix}$ bu vektörler için dik iz düşüm uzunluğunu bularak hacmi bulacağız



$$\frac{\vec{a} \cdot \vec{b}}{|\vec{b}|} \Rightarrow \frac{\begin{pmatrix} 1 \\ -5 \end{pmatrix} \begin{pmatrix} x_2 \\ y_2 \end{pmatrix}}{\left| \begin{pmatrix} 1 \\ -5 \end{pmatrix} \right|} \Rightarrow \frac{x_2 - 5y_2}{\sqrt{1^2 + 5^2}}$$

$$\text{Hacim} = \frac{x_2 - 5y_2}{\sqrt{26}} \cdot \sqrt{26} \cdot \frac{1}{2} = \frac{x_2 - 5y_2}{2}$$

c-) $x + 4y - z = 6$

$x - 5y - 0 = 0$

$$\left[\begin{array}{ccc|c} 1 & 4 & -1 & 6 \\ 1 & -5 & 0 & 0 \end{array}\right] \xrightarrow{-R_1 + R_2 \to R_2} \left[\begin{array}{ccc|c} 1 & 4 & -1 & 6 \\ 0 & -9 & 1 & -6 \end{array}\right]$$

P P F

2 Dimensions inside R^3

CamScanner ile tarandı

3-) $H = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - 2 \cdot \begin{bmatrix} 1 \\ 0 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 \end{bmatrix}$

a-)

$$H = -\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$H^2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

b-) $H^T \cdot H = I$

$-\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \cdot -\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I \rightarrow$ ortogonal

ortonormal için $\frac{\begin{pmatrix} 1 \\ 0 \end{pmatrix}}{\left|\begin{matrix} 1 \\ 0 \end{matrix}\right|}$, $\frac{\begin{matrix} 0 \\ 1 \end{matrix}}{\left|\begin{matrix} 1 \\ 0 \end{matrix}\right|}$

c-) $|H - \lambda I| = 0$ $\begin{bmatrix} 1-\lambda & 0 \\ 0 & 1-\lambda \end{bmatrix}$ → λ lar için determinantı

$1 - \lambda - \lambda + \lambda^2 \Rightarrow \lambda^2 - 2\lambda + 1$ $\quad \lambda = 1$

λ
λ

$\lambda = 1$ için özvektör

$$\begin{bmatrix} -2 & 0 \\ 0 & -2 \end{bmatrix} \cdot \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$-2x_1 = 0 \rightarrow x_2 = a$

$-2x_2 = 0 \quad x_1 = a$

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = a \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

d-) Köşegen hale getirme → 1. öz değerler ($\lambda = 1$)
2. öz vektörler $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$
3. $D = P^{-1} H \cdot P$

$$P = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

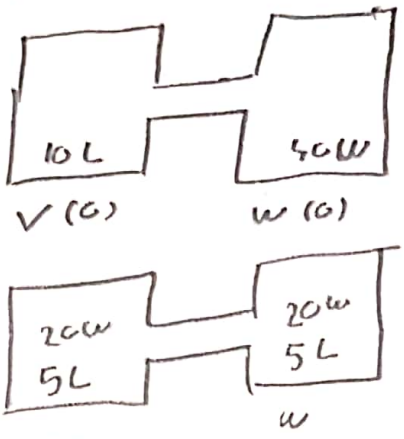
$\det P = 0 \rightarrow$ Köşegenleştirilemez

Boğrum İslamoğlu 150719697

CamScanner ile tarandı

Bayram İslamoğlu 150719697

4-)



a-) $\frac{d}{dt}\begin{bmatrix} v \\ w \end{bmatrix} = \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix}\begin{bmatrix} v \\ w \end{bmatrix}$

b-) %80 w — %20 L

c-) her bir tankta 5 litre sıvı bulunur

1-)

b-) $U = \begin{bmatrix} 1 & c & c & 0 \\ 0 & 1 & c & c \\ 0 & c & 1 & 0 \\ 0 & 1 & -\frac{1}{2} & 1 \end{bmatrix}$

a-) $\xrightarrow[-2R_1+R_3 \to R_3 \\ -R_1+R_4 \to R_4]{-3\alpha R_1+R_2 \to R_2} \begin{bmatrix} 1 & \alpha & -\gamma & \alpha \\ 0 & (5-3\alpha) & (-3\gamma-3\alpha) & (1-3\alpha) \\ 0 & (\beta-2\alpha) & (3+2\gamma) & (\alpha^2-2\alpha) \\ 0 & (14-\alpha) & (-6+\gamma) & (2-\alpha) \end{bmatrix}$

$\alpha = 1 \quad \beta = 1 \quad \gamma = -1 \quad k = 13$

CamScanner ile tarandı